FeaLearner: A Novel Framework of Self-Adaptive Feature Learning and Selection for Suicide Risk Detection from Users' Social Media Posts

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Abstract. Social media posts often reflect users' emotions and thoughts, sometimes containing subtle yet critical signals of suicidal tendencies. Traditional feature engineering-based methods often miss concealed distress. Although time-aware sequence-based methods are effective at learning and aggregating latent features from a user's historical emotional spectrum, they frequently struggle to explicitly capture variations in the degree of suicidality and the presence of suicide-related associations across a user's sequence of posts. A more effective approach would involve a deeper integration of both latent emotional historic context and textual features tailored to the specified user. In this paper, we propose an innovative self-adaptive **Fea**ture **Learn**ing framework (named FeaLearner), which transforms the suicide risk detection task into self-adaptive feature learning and selection procedures: firstly, it leverages a BERT-BiLSTM model to track users' psychological-emotional evolution, enhanced by a temporal attention mechanism to highlight emotional historic context features. Furthermore, a multi-view adaptive feature selection network dynamically weights suicide-related textual features by integrating diverse sub-network perspectives, improving optimal text feature selection. Finally, the framework integrates emotional historic and textual context feature representations, and formulates an ordinal regression problem for suicide risk-level prediction. Experimental results demonstrate that FeaLearner yields better performance compared to various competitive baselines.

Keywords: Self-adaptive feature learning · Multi-view adaptive feature selection network · Users' social media posts

M. Yoshikawa et al. (Eds.): ADMA 2025, LNAI 16198, pp. 207–222, 2026.
https://doi.org/10.1007/978-981-95-3456-2_15

1 Introduction

In recent years, social media platforms such as Weibo and Twitter have become crucial channels for individuals to express their emotions and psychological states. The anonymity afforded by these platforms encourages open participation in online social interactions, particularly among younger demographics. Posts in textual form shared on social media often offer a more authentic reflection of a user's current state of mind and can be contextualized within broader situational frameworks. This makes them a valuable resource for monitoring and analyzing users' mental health and presents novel opportunities for research in the detection of suicidal ideation.

Promisingly, multi-faceted features such as linguistic, syntactic structures, and contextual embedding are effectively combined and fed into classifiers to detect suicide risk [8,14,24]. Despite their efficacy, these feature engineering-based methods predominantly rely on explicit cues present in users' social media content, often failing to identify abnormal psychological states, particularly when users intentionally conceal emotional distress. Moreover, they tend to overlook the influence of different types of features and their intrinsic factors to the posts themselves, thereby limiting the overall performance of suicide risk detection.

Taking into account a user's mental change expressed in their historical posts and temporal post irregularities, the deep learning-based architecture like convolutional neural networks (CNN) [9] and recurrent neural networks (RNN) [6] are employed to automatically learning hidden features on users' historical post sequences for suicide risk detection. Glen *et al.* [3] use an RNN model and attention mechanism to detect a user at risk of suicide or not. Cao *et al.* [19] employ an LSTM and fasttext-based architecture for modelling temporal context. Ramit *et al.* [20] investigate the ability of deep learning architectures to build an accurate and robust model for suicidal ideation detection. On one hand, deep learning methods with attention mechanisms effectively capture implicit suicidal semantics within user-associated posts, which provides us with valuable insights for modeling the user's emotional historic context. On the other hand, due to insufficient emotion expression, these methods still have room by further leveraging the complementary nature of both implicit features from the emotional historic context and explicit multi-faceted textual features to enhance model performance.

Recent advances in suicide risk assessment have demonstrated the importance of fine-grained evaluation for at-risk users, differentiating varying severity levels rather than treating them equally. Drawing inspiration from Ramit *et al.* [19] 's work, who reformulate the suicide task assessment as an ordinal regression problem and abide by the C-SSRS scale [16], where not all wrong classes are equally wrong to prioritize at-risk users based on the increasing order of suicide risk levels. In our work, we similarly adopt this ordinal framework to assess suicide risk levels in users' social media posts.

Based on the aforementioned analysis, we develop an innovative learning framework with significant feature self-adaptive procedures: feature learning and selection, which we call FeaLearner. In more detail, this framework uses a BERT-Bi-LSTM architecture to model the user's psychological and emotional evolution, and highlight the most discriminative features with a temporal attention mechanism in the historical post context. it introduces a multi-view adaptive feature selection network, which computes categories of suicide-related textual feature importance by incorporating various views from multiple sub-networks, promoting a more effective feature selection. Finally, it combines the user's emotional historic context feature with the textual feature representations for suicide risk-level prediction. Our approach enables a more accurate understanding and analysis of users' social media posts, thereby enhancing the performance of the suicidal risk detection task. In summary, our main contributions of this paper can be summarized as follows:

– We propose FeaLearner, an innovative framework that integrates self-adaptive feature learning and selection. By capturing the complementary nature of multi-type features, it better enhances suicide risk detection performance.
– FeaLearner captures users' significant psychological and emotional fluctuations over time by modeling historical post sequences with BERT-BiLSTM architecture and a temporal attention mechanism. Moreover, the framework incorporates a multi-view adaptive feature selection network to select informative suicide-aware textual features while avoiding bias toward feature classes.
– Experimental results on two widely used suicide risk detection datasets (i.e., Weibo and Reddit) demonstrate that FeaLearner achieves comparable performance compared to strong baselines. Further, ablation and effectiveness analyses validate the essential roles of different component in the FeaLearner, especially for the multi-view adaptive feature selection, feature combinations strategy[1].

2 Related Work

2.1 Feature Engineering-Based Methods

Researchers and practitioners can detect early warning signs in user textual context in social media by leveraging explicit features for targeted interventions. Following ideas of feature engineering-based methods, Wang *et al.* [24] study various lexical, syntactic, and knowledge-based features from suicide notes for suicide risk detection. Ji *et al.* [8] analyze online user-generated content and extract a combination of statistical, syntactic, linguistic, word embedding, and topic features to detect suicidal ideation. Mulholland and Quinn *et al.* [14] extract lexical and syntactic features from a corpus of songs to establish a classifier that predicts the likelihood of suicide in songwriters. Huang *et al.* [7] establish a psychological dictionary by expanding "HowNet" and extract N-gram features and sentence sentiment polarity to detect suicidal behavior on Weibo dataset. However, these methods mainly depend on users' explicit outward textual expressions on social media, and fail to infer abnormal states, particularly when users deliberately hide emotional distress. On the other hand, these extracted features are treated uniformly without properly quantifying their importances, limiting the model's detection performance.

[1] To illustrate the evaluation of our proposed FeaLearner framework and facilitate further research on this topic, we have made the experimental details and source code of the framework publicly available at https://github.com/shengyp/FeaLearner.

2.2 Deep Learning-Based Methods

On benifit of modelling ability of deep learning-based methods for temporal sequence context. Tadesse *et al.* [23] built an LSTM-CNN model to detect suicidal ideation based on word2vec features. Sosa *et al.* [22] compare CNN-LSTM and LSTM-CNN models, analyzing the impact of module arrangement order on understanding context. These RNN and LSTM-based approaches assume that users' historical posts are equally spaced in time, hindering the suicide ideation detection model's ability to learn their relative importance in a time-aware manner. Renjith *et al.* [17] introduce an attention mechanism on the basis of combining LSTM with CNN models, focusing on different important features in the input sequence to understand the context in a fine-grained manner for suicide detection. Naseem *et al.* [15] propose a hierarchical attention network based on graphs to capture semantic, syntactic, and sequential contextual information to understand the emotional historic context. In practice, we found deep learning models incorporating attention mechanisms having led to some performance improvements on this task. Hence, we follow Ramit *et al.*'s work [19] and employ a temporal attention mechanism to measure the degree of suicidality and the presence of suicide-related associations across a user's sequence of posts. In addition, we further consider the complementary feature advantages and attempt to combine implicit feature representations derived from the emotional historic context and explicit multi-faceted textual features that represent a user's emotions, language styles, and expressions for suicide risk detection.

3 Methodology

Problem Formulation. Our task aims to assess suicide risk for each user $u_i \in U = \{u_1, u_2, \cdots, u_n\}$ by analyzing their chronologically-ordered social media posts $P_i = \{p_{i,1}, p_{i,2}, \cdots, p_{i,N_i}\}$, where $p_{i,t}$ denotes the t-th post published by u_i. Following the idea of Ramit *et al.* [19], we also formulate suicide risk assessment as an **Ordinal Regression problem**, essentially modeling risk levels as hierarchically ordered categories. This setting accounts for the inherent hierarchy among risk categories while ensuring more nuanced and interpretable predictions.

The Overview Architecture of FeaLearner. We present FeaLearner, a framework that conducts self-adaptive feature learning and selection to enhance the suicide risk detection task. FeaLearner models a user's historical post sequence and forms a corresponding contextual representation (Sect. 3.1), designs a multi-view adaptive feature selection network, which computes categories of suicide-related feature importance by incorporating various views from multiple sub-networks (Sect. 3.2), and combines the user's emotional historic contextual feature representation and its suicide-aware textual feature representation for suicide risk-level prediction (Sect. 3.3). The overall architecture is illustrated in Fig. 1.

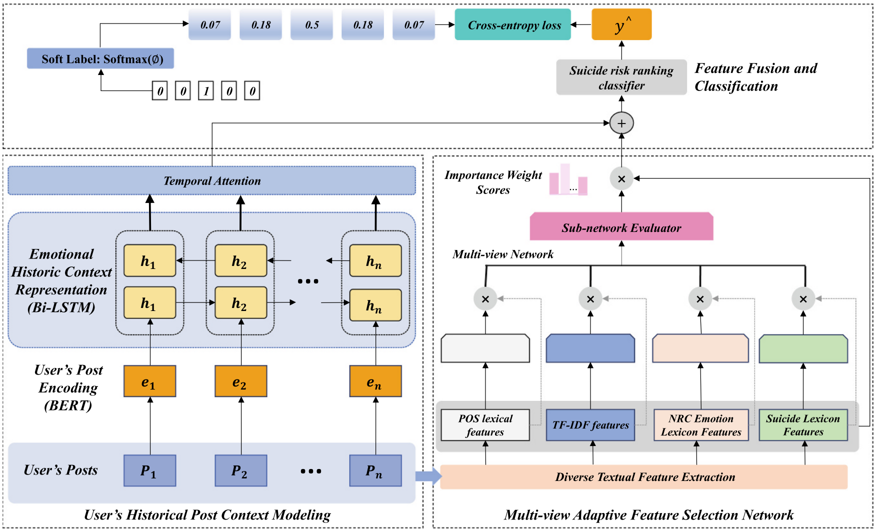

Fig. 1. An overview of the proposed FeaLearner framework. (1) On the left side, we use BERT for the post encoding, followed by Bi-LSTM to model psychological and emotional evolution. An attention mechanism adaptively weights posts, emphasizing those with high emotional volatility or strong suicide risk correlation. (2) On the right side, we introduce a multi-view adaptive feature selection network consisting of multiple sub-networks, each of which adaptively learns to measure the feature importance of each type of suicide-related textual feature and its internal elements for the feature selection. (3) We integrate the user's emotional historic context feature with suicide-aware textual feature representations, employing an ordinal regression/cross-entropy loss for final suicide risk-level prediction.

3.1 User's Historical Post Context Modeling

User's Post Encoding. Each user's post may contain indicative and potential suicide risk signals, offering valuable insights into their mental state and helping build a comprehensive understanding of their psychological well-being [19]. Building on advances in transfer learning and pre-trained language models for NLP tasks, we leverage a popular pre-trained BERT model [4] for post-level contextual representation. Each post $p_{i,t}$ of user i can be encoded as $\mathbf{e}_{i,t} = \mathrm{BERT}(p_{i,t})$.

Emotional Historic Context Representation of Posts. Analyzing the sequence of a user's historical posts can track evolving psychological states and risk levels over time. The Bi-LSTM network's enhanced ability to capture long-term dependencies makes it particularly suitable for modeling these emotional and psychological shifts across multiple posts. For each encoded post $\mathbf{e}_{i,t}$ in user i's historical post sequence, it can obtain its contextual representation by harvesing context from historic as well as future posts, i.e., concatenating the left-to-right and the right-to-left hidden state vectors of the LSTM is as follows:

$$\overrightarrow{\mathbf{h}_{i,t}} = \mathrm{LSTM}(\mathbf{e}_{i,t}, \overrightarrow{\mathbf{h}_{i,t-1}}), \overleftarrow{\mathbf{h}_{i,t}} = \mathrm{LSTM}(\mathbf{e}_{i,t}, \overleftarrow{\mathbf{h}_{i,t-1}}), \mathbf{h}_{i,t} = [\overrightarrow{\mathbf{h}_{i,t}}, \overleftarrow{\mathbf{h}_{i,t}}]. \quad (1)$$

The Bi-LSMT transform user i's historical post encodings $\mathbf{E}_i = \{\mathbf{e}_{i,1}, \mathbf{e}_{i,2}, \ldots, \mathbf{e}_{i,N_i}\}$ into contextual representations $\mathbf{H}_i = \{\mathbf{h}_{i,1}, \mathbf{h}_{i,2}, \ldots, \mathbf{h}_{i,N_i}\} \in \mathbb{R}^{d \times N_i}$, where d is the dimension of each hidden state vector. N_i is the number of historical posts published by user i.

Temporal Attention. Due to the severity of suicide risk and the presence of suicidal content vary across each user's posts. Inspired by Ramit *et al.* [19], we introduce a temporal attention mechanism to automatically assign higher weights to user i's posts that are more indicative of higher risk and aggregate them, forming the final emotional historic contextual representation of posts, formally described as:

$$\tilde{\alpha}_{i,t} = \mathbf{c}^T tanh(\mathbf{W}_x \mathbf{h}_{i,t} + \mathbf{b}_x), \alpha_{i,t} = \frac{exp(\tilde{\alpha}_{i,t})}{\sum_{t=1}^{T_i} exp(\tilde{\alpha}_{i,t})}, \mathbf{E}_{i,his} = \sum_{t=1}^{N_i} \alpha_{i,t} \mathbf{h}_{i,t}, \quad (2)$$

where $\mathbf{W}_x \in \mathbb{R}^{d \times d}$ and $\mathbf{c}^T \in \mathbb{R}^d$ are the learned network parameters, $\tilde{\alpha}_{i,t}$ is the attention weight for the t-th historical post of user i.

3.2 Multi-view Adaptive Feature Selection Network

Diverse Textual Feature Extraction. Through in-depth analysis of the data, we extract four types of suicide-related textual features from user' social media posts: POS lexical features[2], the TF-IDF features [18], NRC Emotion Lexicon Features (ELF) [12,13], and Suicide Lexicon Features (SLF) [10]. These features could indicate suicidal risk, and their analysis may reveal deeper insights about the user's mental state.

Multi-view Network. Inspired by the mixture-of-experts model [21], we design a multi-view network comprising K sub-networks $(SN_1, SN_2, \ldots, SN_k)$ that quantify the importance of each class of suicide-related textual features and its internal feature elements (i.g., words) . Each feature class associated with the sub-network is defined as $\tilde{\mathbf{F}}_\mathbf{i} = [\tilde{f_{i,1}}, \tilde{f_{i,2}}, \ldots, \tilde{f_{i,n}}]$ represents the i-th textual feature type, where $\tilde{f_{i,n}}$ is the n-th internal feature element. We calculate the influence of internal feature elements in the i-th feature class as follows:

$$\mathbf{SN}_i = Relu(\tilde{\mathbf{F}}_i \mathbf{W}_i + \mathbf{b}_i) = \left[SN_{i,1}, SN_{i,2} \ldots, SN_{i,n}\right] (i < K), \quad (3)$$

where $\mathbf{W}_i$ and $\mathbf{b}_i$ are trainable network parameters in the i-th sub-network, $Relu$ is the activation function.

We obtain a representation of the i-th type of textual feature as follows:

$$\mathbf{F}_i = \tilde{\mathbf{F}}_i \odot \mathbf{SN}_i = \left[\tilde{f_{i,1}} \cdot SN_{i,1}, \tilde{f_{i,2}} \cdot SN_{i,2}, \ldots, \tilde{f_{i,n}} \cdot SN_{i,n}\right], \quad (4)$$

where $\mathbf{F}_i = \left[f_{i,1}, f_{i,2}, \ldots, f_{i,n}\right]$ represents feature vector corresponds to i-th feature class.

[2] We employ the Jieba segmentation tool (https://github.com/fxsjy/jieba) for part-of-speech tagging on the Weibo dataset and utilized the NLTK tokenizer (https://www.nltk.org) for POS tagging on the Reddit dataset. This enables us to statistically analyze the frequency of each part-of-speech category in users' posts, which served as our Part-of-Speech (POS) lexical features.

Sub-network Evaluator. Analyzing the importance of each type of textual feature in detecting the user's suicide risk and assigning different weights to them is crucial for highlighting the features that play key indicative roles in the detection task. We achieve an adaptive feature selection strategy by multiplying user i' feature vectors $\mathbf{F}_{i,sn} = [\mathbf{F}_{i,1}, \mathbf{F}_{i,2}, \ldots, \mathbf{F}_{i,K}]$ with the corresponding importance weight scores $\mathbf{G}_{i,sn} = [\mathbf{G}_{i,1}, \mathbf{G}_{i,2}, \ldots, \mathbf{G}_{i,K}]$. The process is formalized as:

$$\mathbf{G}_{i,sn} = Relu(\mathbf{W}_g \mathbf{F}_{i,sn} + \mathbf{b}_g) = [g_{i,1}, g_{i,2}, \ldots, g_{i,K}] \tag{5}$$

$$\mathbf{T}_i = \mathbf{G}_{i,sn} \odot \mathbf{F}_{i,sn} = \left[g_{i,1} \cdot \mathbf{F}_{i,1}, g_{i,2} \cdot \mathbf{F}_{i,2}, \ldots, g_{i,K} \cdot \mathbf{F}_{i,K}\right], \tag{6}$$

where $\mathbf{W}_g$ and $\mathbf{b}_g$ are the shared trainable parameters for each sub-network, and $Relu$ is the activation function. $\mathbf{T}_i$ denotes user i's final textual feature representation.

3.3 Feature Fusion and Classification

Feature Fusion. We combine user i' emotional historic context feature representation $\mathbf{E}_{i,his}$ and its suicide-aware textual feature representation $\mathbf{T}_i$ to form fused feature representation $\mathbf{X}_{i,fuse}$, which is formally presented as follows:

$$\mathbf{X}_{i,fuse} = [\mathbf{E}_{i,his}; \mathbf{T}_i]. \tag{7}$$

We adopt a two-layer fully connected network as the user's suicide risk ranking classifier. The first fully connected layer applies a linear transformation to the fused feature $\mathbf{X}_{i,fuse}$, and fed into the second fully connected layer yields the classification scores $\hat{\mathbf{y}}$ as follows:

$$\mathbf{h}_1 = f_1(\mathbf{X}_{i,fuse}) = \mathbf{W}_1 Dropout(\mathbf{X}_{i,fuse}) + \mathbf{b}_1, \tag{8}$$

$$\hat{\mathbf{y}} = f_2(\mathbf{h}_1) = \mathbf{W}_2 \mathbf{h}_1 + \mathbf{b_2}, \tag{9}$$

where $\mathbf{W}_1$, $\mathbf{b}_1$ are the learned network parameters in the first fully connected layer, $\mathbf{W}_2$, $\mathbf{b}_2$ are the learned network parameters in the second fully connected layer.

Ordinal Regression. For the Reddit dataset, inspired by the work of Tadesse *et al.* [23], we train our model by minimizing an ordinal regression loss. We define the set $Y = \{\text{SU} = 0, \text{IN} = 1, \text{ID} = 2, \text{BR} = 3, \text{AT} = 4\} = \{r_i\}_{i=0}^{4}$ to represent the five ordinal risk-levels in the increasing order of severity. Given a specified actual risk-level $r_t \in Y$, we compute soft-encoding labels as probability distributions $\mathbf{y} = [y_0, y_1, \ldots, y_4]$ of ground truth labels. The probability values y_i of each risk-level r_i is calculated as follows:

$$y_i = \frac{exp^{-\phi(r_t, r_i)}}{\sum_{k=1}^{\lambda} exp^{-\phi(r_t, r_i)}} \forall r_i \in Y. \tag{10}$$

Here $\phi(r_t, r_i) = \beta |r_t - r_i|$ is the cost function used to penalize how far the actual risk-level r_t is from the predicted risk-level r_i, and β is the parameter used to adjust the magnitude of penalty for incorrectly predicted risk-level.

After computing the probability distribution $\mathbf{y}$ for the ground truth label, we ultimately use the classification confidence scores to calculate the cross-entropy loss as:

$$\mathcal{L} = -\frac{1}{n}\sum_{j=1}^{n}\sum_{i=1}^{\lambda}\mathbf{y}_{ij}\log\hat{\mathbf{y}}_{ij}, \tag{11}$$

where n represents the batch size and λ represents the number of risk-levels (5 ranking in total).

For the Weibo dataset, we employ the cross-entropy loss function due to its balanced binary class distribution defined as follows:

$$\mathcal{L} = -\frac{1}{N}\sum_{i=1}^{N}\mathbf{y}_i\log\hat{\mathbf{y}}_i + (1-\mathbf{y}_i)\log(1-\hat{\mathbf{y}}_i), \tag{12}$$

where N is the number of samples, $\mathbf{y}_i$ denote the ground-truth probability values of risk-level, and $\hat{\mathbf{y}}$ represent the predicted probability values of risk-level.

4 Experiments and Analysis

4.1 Datasets and Evaluation Metrics

Datasets. We evaluate our model and baselines on two popular datasets: Weibo and Reddit datasets. In the following, we detail each dataset.

- **Weibo dataset.** This dataset was firstly collected by Cao *et al.* [2] from Weibo. On the benifit of the expert's analysis, 3,652 users were retained, all of whom had explicitly mentioned suicidal thoughts, suicide plans, or self-harm behaviors at least five times on different days. As a control group, they selected users who had never posted any content related to suicidal ideation and labeled them as non-suicidal, ultimately including 3,677 ordinary users as the study sample. In our experiments, we preprocessed and split the dataset into training (5,785 samples), validation (723 samples), and test sets (723 samples), with balanced suicidal (labeled as 1) and non-suicidal (labeled as 0) samples.
- **Reddit dataset.** This dataset was refined by Ramit *et al.* [19], which consists of reddit posts of 500 users across 9 mental health and suicide related subreddits. It adopts the Columbia-Suicide Severity Rating Scale (C-SSRS) [16] to categorize suicide risk into five ordinal levels of increasing severity: Support (SU), Indication (IN), Ideation (ID), Behavior (BR), and Attempt (AT)[3]. In our experiments, we preprocessed and split the dataset into training (303 samples), validation (99 samples), and test sets (98 samples), with the Class-wise distribution: 20% SU, 20% IN, 34% ID, 15% BR, and 9% AT.

[3] In the C-SSRS, users are classified into five ascending risk levels: Support (SU) $<$ Ideation (ID) $<$ Behavior (BR) $<$ Attempt (AT), reflecting increasing suicide risk severity.

Evaluation Metrics. Following current mainstream evaluation metrics for the task [5, 19], we use Accuracy (Acc), F1-score (F1), GP (Graded Precision), GR (Graded Recall), and F-score to assess the extent to which model predictions deviate from ground truth labels[4]. False Negative (FN) is refined as the proportion of number of times predicted severity of suicide risk level k^p is less than the actual risk level k^α, and False Positive (FP) is the proportion of number of times the predicted risk k^p is greater than the actual risk k^α. Precision and recall are reformulated as GP and GR to incorporate the ordering among risk levels.

$$FN = \frac{\sum_{i=1}^{N_T} I(k_i^a > k_i^p)}{N_T}, FP = \frac{\sum_{i=1}^{N_T} I(k_i^p > k_i^a)}{N_T} \tag{13}$$

$$GR = \frac{TP}{TP + FN}, GP = \frac{TP}{TP + FP}, F - score = \frac{2GR \times GP}{GR + GP}, \tag{14}$$

where I is an indicator function that equals 1 if the condition holds and 0 otherwise. N_T represents the size of the test sets.

4.2 Baselines and Implementation Details

Baselines for Comparison. We compare our FeaLearner framework with three categories of SRD models: (1) *Feature engineering-based methods*, including SVM + RBF [1]. (2) *Deep learning-base methods*, including Contextual CNN [5], ContextBERT [11], LSTM+Attention. (3) *Sequence-based methods*, including SDM [2] and SISMO [19].

Implementation Details. The Bi-LSTM layer consisted of 128 neurons, while the pre-trained model generated embeddings with a dimensionality of 768. The learning rate is initialized at 0.01, and the Adam optimizer is employed for training. Additionally, the parameter β is set to 2 to balance the loss components. All of the experiments are processed on a Linux server with CPU Xeon Gold 6142, RAM 64G, and Nvidia 4090 GPU. All methods are implemented using Pytorch platform[5].

4.3 Model Overall Performance Results

We report overall performance results of the FeaLearner model and baseline methods in Table 1. The major observations from the results can be summarized as follows: (1) Deep learning-based methods outperform traditional feature engineering (e.g., SVM) by better capturing contextual semantics and emotional states in posts, enabling more effective suicide risk detection. In contrast, feature engineering-based approaches often

[4] For the binary class-balanced Weibo dataset, we evaluate our model's performance using Accuracy (correct classification ratio) and F1-score (Precision-Recall harmonic mean). For the Reddit dataset, the GP metric is to evaluate high-risk prediction accuracy, while the GR metric is to assess comprehensive high-risk case identification, enabling more effective detection of high-risk populations. The F-score not only prevents overemphasis on either suicide risk-level but also better adapts to this class-imbalanced dataset.

[5] https://pytorch.org/.

rely on explicit textual features, which often fail to adequately detect abnormal states, especially when users intentionally conceal their emotional distress, limiting their effectiveness; (2) Sequence-based methods, such as SDM, ContextBERT perform better than the contextual CNN, we postulate this advantage in learning better representations from a user's evolving emotional states over time in comparison to the Contextual CNN's modeling ability of local features in the temporal context; (3) FeaLearner achieves the best performance across the majority of metrics. Moreover, we analyze the percentage of correctly predicted and mispredicted categories for risk-level on the ground-truth labels, as shown in Fig. 2, it outperforms strong baseline methods (e.g., Contextual CNN (shortened as C-CNN), ContextBERT (shortened as C-BERT), SISMO), and this can be credited to the following factors: (a) Our model not only effectively captures emotional historic context features in the user's historical post sequence, but also adaptively learns and selects salient textual contextual features. Their integration fully leverages the complementary strengths of both feature types. In contrast, baseline models only address one aspect of the aforementioned modeling framework. (b) Ordered regression loss employed in our model is able to effectively penalize the deviation of the model in predicting the risk level concerning the actual risk level on the C-SSRS scale, as opposed to the baselines, such as Contextual CNN, ContextBERT, which treat each misclassification equally.

Table 1. Experimental results of user suicide risk detection on Weibo and Reddit datasets. The best-performing results for each evaluation metric are highlighted in bold, while the second-best results are marked with an underline, respectively.

Model	**Weibo**		**Reddit**		
	Acc (%)	F1 (%)	GP	GR	F-score
SVM+RBF	70.34	69.01	0.53	0.51	0.52
Contextual CNN	86.77	84.47	<u>0.69</u>	0.52	0.59
ContextBERT (Text)	87.41	87.40	0.63	0.57	0.60
LSTM+Attention	88.89	88.10	0.67	0.50	0.57
SDM (Text)	88.56	87.99	0.60	0.54	0.57
SISMO	<u>89.07</u>	<u>89.07</u>	0.66	**0.61**	<u>0.64</u>
FeaLearner (Ours)	**91.02**	**91.08**	**0.77**	<u>0.73</u>	**0.75**

4.4 Model Ablation and Effectiveness Analysis

Ablation Study. We analyze the performance of the FeaLearner framework by gradually adding emotional historic context features (EHCF), suicide-related textual features (SRTF), and multi-view adaptive feature selection (MVAFS) components, as described in Table 2. The experimental results show that the combination of EHCF and SRTF can achieve complementary enhancement of suicide risk signals, thus improving the model performance and verifying the indispensability of both types of information in

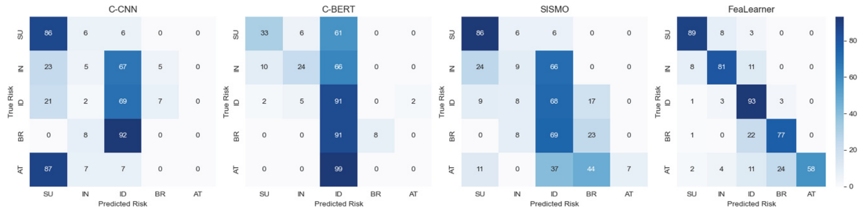

Fig. 2. Normalized confusion matrix for prediction results. A row in each figure represents examples of actual risk-level, the columns represent predicted risk levels. The values of diagonal elements represent the percentage of correct predictions.

risk prediction tasks. Notably, after adding the MVAFS component as FeaLearner, it can analyze the importance of suicide-related textual features in a more detailed manner, effectively identify feature information with key predictive value, and further enhances the overall prediction capability.

Table 2. Ablation results over FeaLearner's components.

	Weibo		Reddit		
	Acc (%)	F1 (%)	GP	GR	F-score
EHCF	88.95	88.79	0.70	0.51	0.59
SRTF	80.73	80.94	0.67	0.59	0.63
EHCF+SRTF	90.33	90.22	0.74	0.55	0.63
FeaLearner (ours)	**91.02**	**91.08**	**0.77**	**0.73**	**0.75**

Analysis of Multi-view Adaptive Feature Selection. To validate the effectiveness of the introduced multi-view adaptive feature selection network, we visualize and analyze the importance weights of adaptive selection across types of textual and their internal elements features. It can be divided into two parts:

(1) Multi-view network. We calculate the sum of the weights of the internal feature elements of the multi-type textual features after importance analysis and divide them by the total number of sample instances. The corresponding heatmaps are shown in Figs. 3 and 4. We present the Top-10 feature elements with the highest computed importance weights, ranked in descending order. It can be observed that sub-networks in the multi-view network effectively assign varying levels of importance to internal feature elements within each type of textual feature. For example, for the POS lexical features, the Reddit dataset places greater emphasis on categories such as CC (coordinating conjunction), JJ (adjective), and PRP (personal pronoun). While for the Suicide Lexicon Features (SLF), the Reddit dataset focuses more on low-risk terms, the Weibo dataset prioritizes vocabulary related to mental illness, suicidal ideation, and discussions about others.

Fig. 3. Visualization of adaptively selected suicide-related textual feature importances by the multi-view network on the Weibo dataset.

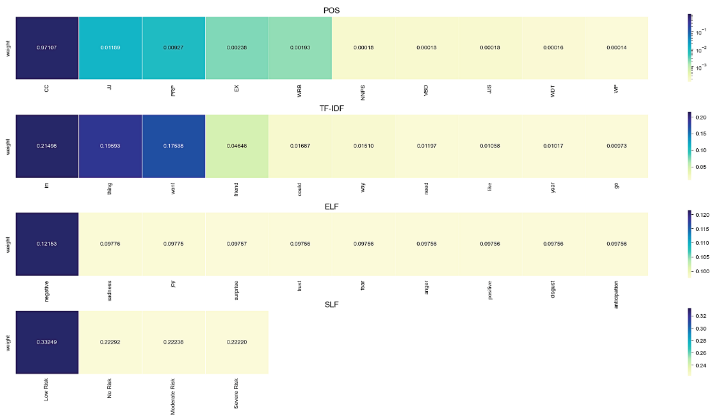

Fig. 4. Visualization of adaptively selected suicide-related textual feature importances by the multi-view network on the Reddit dataset.

(2) Sub-network evaluator. Analyze the importance of each type of textual feature, and the importance weight heatmap is shown in Fig. 5. The results indicate that both datasets pay relatively few attention to POS part-of-speech features and tend to rely more on TF-IDF features. This is because users often employ informal language on social media, and POS features perform relatively weakly in capturing emotional states and signals of suicidal tendencies. In contrast, TF-IDF features, by analyzing the frequency and distribution of keywords in the text, can effectively identify the themes and critical content of the text, thus carrying higher weight in suicide tendency detection. Additionally, there are some differences in the weight allocation for NRC Emotion Lexicon Features (ELF) and Suicide Lexicon Features (SLF) across the datasets. The Reddit dataset assigns similar importance weights to both ELF and SLF, whereas the Weibo dataset allocates a higher weight to SLF. This is because the emotional words in

the ELF are constructed based on English, and their relevance may diminish in the context of Chinese social media due to differences in cultural background and emotional expression. In comparison, the Chinese suicide lexicon is meticulously designed based on the linguistic environment and cultural context of Chinese social media platforms like Weibo. Its vocabulary is more directly and accurately associated with suicidal tendencies, hence its higher importance.

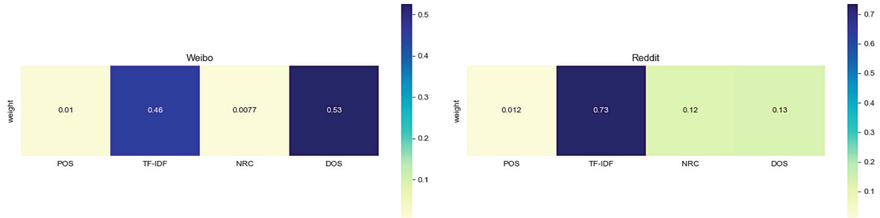

Fig. 5. Visualization of textual feature importances by category through the sub-network evaluator on Weibo and Reddit datasets.

Analysis of Suicide-Aware Feature Combination Strategy. We employ three distinct machine learning models along with our proposed FeaLearner model to investigate how progressively incorporating different categories of textual features affects suicide risk prediction performance. As depicted in the Table 3, after adding POS lexical features, TF-IDF, ELF, and SLF sequentially to the XGBoost model, the F1 increases from 80.53% to 83.02% in the Weibo dataset, while the F-score improves from 47% to 51%, and the GP improves from 0.52 to 0.56 in the Reddit dataset. An interesting finding is that model performance does not always improve during the sequential combination of multiple textual features. This is because while feature combinations help highlight suicide-related information, too many redundant features may instead interfere with model judgments. On the other hand, the importance of different types of textual features and their internal elements for suicidal tendency detection varies, further validating the need for adaptive feature importance analysis in our designed multi-view network.

5 Discussion

Ethical Considerations. Our work utilizes datasets from users' social media posts solely for research purposes, with no modification or interference to users' original data. We explicitly emphasize that our model is designed exclusively for suicide risk assessment and possesses no clinical diagnostic capability.

Limitations. Assessing suicide risk through the textual content of social media posts involves inherent subjectivity, which may lead to inconsistencies in evaluations by different occupational psychologists or clinicians. Suicidal ideation is a complex phenomenon shaped by the interplay of multiple factors, and labeled datasets often fail

Table 3. Performance comparison of different models across diverse feature combinations

Models	Feature Combination	Weibo		Reddit		
		Acc (%)	F1 (%)	GP	GR	F-score
LR	POS	74.46	73.66	0.49	0.44	0.46
	POS+TF-IDF	77.98	77.23	0.50	0.44	0.47
	POS+TF-IDF+ELF	79.64	78.96	0.51	0.45	0.48
	POS+TF-IDF+ELF+SLF	79.61	79.00	0.52	0.45	0.48
SVM	POS	76.25	74.53	0.47	0.53	0.50
	POS+TF-IDF	80.16	79.73	0.49	0.54	0.51
	POS+TF-IDF+ELF	81.47	81.09	0.49	**0.55**	**0.52**
	POS+TF-IDF+ELF+SLF	80.54	80.01	0.48	0.54	0.51
XGBoost	POS	80.68	80.53	0.52	0.44	0.47
	POS+TF-IDF	82.16	81.98	0.55	0.44	0.49
	POS+TF-IDF+ELF	83.13	82.88	0.56	0.45	0.50
	POS+TF-IDF+ELF+SLF	83.13	83.02	0.56	0.47	0.51
FeaLearner (ours)	POS+EHCF	90.05	90.11	0.63	0.59	0.61
	POS+TF-IDF+EHCF	90.47	90.56	0.63	0.49	0.55
	POS+TF-IDF+ELF+EHCF	83.13	82.88	0.56	0.45	0.50
	All	**91.02**	**91.08**	**0.77**	**0.73**	**0.75**

to capture the full spectrum of real-world scenarios, introducing potential biases. These biases can stem from variables such as social environment, life stressors, the subjective interpretations of expert annotators, or even inherent platform-specific biases on social media. Consequently, when such analyses are applied in real-life settings, discrepancies in detection outcomes may arise. It is important to emphasize that risk assessments based on social media text do not equate to clinical diagnosis, they serve only as preliminary evaluations. However, with access to more comprehensive contextual data, such as additional user history or behavioral patterns, the more precise detection and timely clinical interventions become achievable.

6 Conclusion and Future Work

We present FeaLearner, a novel self-adaptive feature learning framework that synergistically combines emotional historic context and textual features and selection to enhance the suicide risk detection task. The framework employs: (1) a BERT-BiLSTM architecture with temporal attention to capture important psychological-emotional dynamics from users' historical posts, (2) a multi-view adaptive feature selection network that measures and selects suicide-relevant textual features while maintaining feature type-agnostic bias mitigation, and (3) an ordinal regression loss function for penalizing misclassification for risk levels of different severity. Experimental results show the superiority of the FeaLearner framework compared with several baselines, and the crucial roles of key components are confirmed in the model ablation and effectiveness analysis.

Future research will focus on: (1) Exploring more effective time-aware sequence-based modelling methods to capture the user's emotional commonalities and fluctuations across time in their published posts. (2) Learning the user's fine-grained characteristic features to assist at-risk user detection. (3) Incorporating the user's additional information (e.g., social network relationships, microexpressions) into the model to enhance the suicide risk detection task performance.

Acknowledgements. This work was supported by the National Natural Science Foundation of China (Grant No. 62202075, 62566007), the Natural Science Foundation of Chongqing, China (No. CSTB2022N SCQ-MSX1404), and Fundamental Research Funds for the Central Universities (No. SWU-KR24008). Key Laboratory of Data Science and Smart Education, Hainan Normal University, Ministry of Education (No. DSIE202206), and the State Key Laboratory of Public Big Data, Guizhou University (No. PBD2024-0501).

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